

Project Title

Preventing Wrong Decisions in Smart Building Systems

Mentor Name

Dr. Amit Gaurang

S.No	Rollnumber	Branch	Name	Role	Signature
1.	R2142232076	B.Tech CSE/CSF	Rishi Kumar	Architecture	
2.	R2142230443	B.Tech CSE/CSF	Pramav Venkatraman	IOT Integration	
3.	R2142231346	B.Tech CSE/CSF	Aman Rajpoot	Attack Simulation and Defense	
4.	R2142231273	B.Tech CSE/CSF	Amalkrishnan Ajay	Python Encoder & ML	

23/01/2026

Project Mentor

Cluster Head

Date	Understanding of Project	Project Working	Soft Skills	Report	Mentor Marks	Total Marks	Project Status
R.No	20 Marks	20 Marks	10 Marks	10 MARKS	60 MARKS	100 MARKS	Activity Coordinator

 Synopsis Evaluation
 Theoretical Understanding

Rollno	Problem(4 Marks)	Algorithm(4 Marks)	Data /Data structure(4 Marks)	SWOT Analysis(4 Marks)	Area of Application(4 Marks)	Total Marks(20)

Panel Remark

Reviewer 1 Reviewer 2 Reviewer 3 Reviewer 4 Reviewer 5

 Mid- Term Evaluation
 DESIGN & DEVELOPMENT

Rollno	Technical Diagram(5 Marks)	Programming Concepts(5 Marks)	IPC(5 Marks)	Libraries(5 Marks)	SRS(10)	Total(20 Marks)

Panel Remark

Reviewer 1 Reviewer 2 Reviewer 3 Reviewer 4 Reviewer 5

 End-Term Evaluation
 Testing & Implementation

Rollno	Theoretical Knowledge(5)	Computational Knowledge(5)	Test Case (10)	Soft Skills (10)	Report(5)	Core Computational Skills(15)	Total (50)

Panel Remark

Reviewer 1 Reviewer 2 Reviewer 3 Reviewer 4 Reviewer 5



UNIVERSITY WITH A PURPOSE
Branch

School Of Computer Science
University Of Petroleum and Energy Studies
P.O. Bidholi, Via-Prem Nagar DEHRADUN-248007

Bachelors of Technology in Computer Science Engineering

Minor ☐ Major ☐

Project Title	PREVENTING WRONG DECISIONS IN SMART BUILDINGS SYSTEMS										Mentor Name	Dr. Amit Garg																																																													
Abstract	Smart building systems rely on sensors to automate decisions but often trust incorrect data without verification. Small sensor errors can silently cause energy waste and discomfort without triggering alarms. This project detects such issues by comparing multiple sensor readings and identifying inconsistent behavior. The approach is low-cost, explainable, and compatible with existing systems.																																																																								
Objective	The goal is to prevent wrong automation decisions caused by faulty or manipulated sensor data. The system verifies sensor reliability using majority agreement instead of single readings. It detects silent sensor drift and believable errors. This ensures safer and more efficient building operation.																																																																								
Methodology	Multiple ESP32 sensors monitor the same environment and send data through MQTT. A central system compares readings to form a consensus baseline. Sensors that deviate over time lose trust and are flagged. Decisions are accepted only when most trusted sensors agree.																																																																								
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Guideline: 1) A project group can be of maximum 4 members and no alteration in the group member will be entertained later.
 Guideline: 2) Methodology should have following steps Step1: Literature Review; Step2: Identification Of Requirement (Type Of Data source, Amount Of Data, & Format of Data); Step3: Identification of Algorithm; Step4: Comparative study; Step5: Design and Development of System/Architecture; Step 6: Implementation; Step7: Results
 Guideline:3) Student should upload softcopies of all the documents (reports and power point presentations) in "Project Directory", 24 hrs prior to evaluation.
 Guideline:4) Panel member will give feedback to individual on the scale of 1 to 5 and this scale will change for defaulter i.e. 1 to 3 scale.
 1: Poor 2: Average 3: Good 4: Excellent 5: Outstanding